The Centre for Materials for Electronics Technology (C-MET) is a premier research centre in Electronic materials under Ministry of Electronics and Information Technology (MeitY), Government of India. C-MET is engaged in the development of process/product/technology and transfer of the technology in the area of Electronic materials, components and devices to cater to India's strategic and industrial applications, exploiting indigenous resources of raw materials.

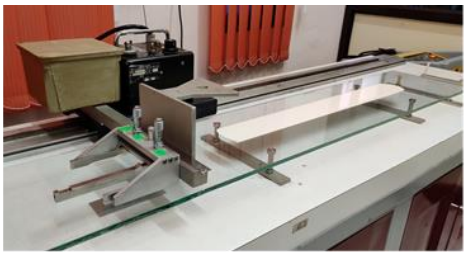
The C-MET family comprises three Centres, located at Pune, Hyderabad and Thrissur with specialized research mandate at each place. C-MET, Thrissur is concentrating on the development of electronic ceramic materials and devices. Its main areas of focus include Clean energy materials, supercapacitors, nano-materials, nano-porous materials, sensors and actuators, microwave materials & nano-structured electronics.

ABOUT THE PRODUCT:

C-MET Thrissur has developed an indigenous spin off technology for Yttria Stabilized Zirconia (YSZ) which are useful for oxygen sensor applications. The probe of the commercial oxygen sensor available in the market has a sensing element which is usually of Zirconia based thin ceramic that forms the heart of the oxygen sensor.

In this ToT, C-MET Thrissur has developed YSZ thin ceramics through tape casting and demonstrated the suitability for oxygen sensor applications. Tape casting is an industrially scalable and cost-effective technique, which has several advantages in terms of tuning the tape dimensions of choice. The YSZ tapes were initially developed for DRDO for high altitude aircraft oxygen sensors and were tested at DEBEL (DRDO) Bangalore.

The performance results were found to be satisfactorily for the oxygen sensors compared to that of the tapes from the imported source. The specifications of the tapes formulated through a 50g batch is mentioned below in table.



3YSZ Ceramic tapes through tape casting



Free standing 3YSZ Ceramic tapes

SPECIFICATIONS OF THE PRODUCT:

SI.No	Specifications	
01	Material Composition	3 Mol% Yttria Stabilized Zirconia (YSZ) powder (1-2 µm Particle size)
02	Tape Dimensions	length: 150 cm (± 1 cm) Width: 10 cm (± 0.1 cm)
03	Thickness range	100 to 200 µm (± 3 µm)
04	Defects	Nearly defect free under optical illumination
05	Green density (dimensional)	≥63% (±1)

06	Sintered Density (Archimedes)	≥98% (±1)
07	Tape formulation	50 g Batch size (can be extended to large scale)

TECHNOLOGY TRANSFER:

C-MET, Thrissur would like to invite competent industrial partners for commercialization of the above listed product through transfer of technology. The technology will be transferred on nonexclusive basis. The ToT package contains the following

1. Document for Tape formulation and Tape casting
2. Processing details
3. Product specification
4. Test procedure
5. Training

THE LEVEL OF TECHNOLOGY READINESS AT C-MET:

- ✓ The technology readiness level is 4

PERIOD OF TRAINING UNDER THE AGREEMENT

- ✓ C-MET, Thrissur will give 3 weeks training to the engineers of the ToT partner at the premises of C-MET.
- ✓ 3 batches of demonstration of nearly defect free YSZ ceramic tapes through tape casting will be provided.

Who can apply?

- ✓ Industries/ start-ups can approach C-MET, Thrissur for the technology. Professionally managed Companies and Corporates are also welcome to apply for the technology.
- ✓ A Committee of experts constituted by the C-MET will assess capabilities and strengths of the industry before finalizing the technology partners.

How to apply?

- ✓ Interested companies may send expression of interest with their details by filling the following questionnaire along with supporting documents to:
  Programme Co-ordinator
  Centre for Materials for Electronics Technology (C-MET)
  Panchawati, Off Pashan Road, Pune:411008
  Email: prog.coord@cmet.gov.in

GENERAL:

- ✓ The industry willing to take technology for commercial exploitation will be required to enter into a ToT agreement with C-MET, Thrissur as per the terms and conditions approved by the competent authority in the MeitY prescribed format.